Fake News Detection Using Supervised Machine Learning.

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Abstract

In the current digital age, deliberate misleading information spread via traditional media channels or online social media platforms is known as fake news. The quick transmission of false information has become an urgent concern as social media is the primary news source for most members of the public due to its ease of use. This study examines the spread of false information through media. It also looks into how it affects the reliability of sources, and why proactive detection techniques are so important. A multifaceted approach utilizing machine learning models is suggested. Text preparation procedures are employed when datasets from several sources are gathered and preprocessed for the study. The study highlights the significance of continuous evaluation and enhancement of models. This helps in adjusting to changing trends and lessens the adverse consequences of false information.

Introduction

When purposefully misleading information or hoaxes are disseminated through traditional media channels or online social media platforms, they are referred to as fake news. Information travels fast in the modern digital age thanks to a multitude of online resources. The ability to spot bogus news has become crucial. Nowadays, a sizable portion of the populace mostly uses social media for news consumption. The proliferation of social networking sites and online news portals is the cause of this. Compared to traditional media outlets like television or newspapers, news on social media is more readily available and less expensive. Consumption of social media has increased as a result of this. Using social media as a news source has grown in popularity due to the simplicity with which friends and the community may provide news.

The characteristics of social media platforms are closely linked to the emergence of fake news. Global, real-time interactions are abundant on social media. Nevertheless, due to its openness and accessibility, misinformation can spread quickly on these platforms (Kudarvalli et al., 2020). Research has indicated that on social media, false information usually performs better than accurate information. This poses a significant challenge to maintaining the news ecosystem's authenticity equilibrium. A proactive approach to identification is necessary due to the proliferation of misleading material on social media. On Twitter, there are no simple ways to report misleading information. However, certain social networking platforms do offer this feature. Scientists are currently exploring the use of machine learning algorithms and news keywords to extract data in real-time and distinguish between genuine and fraudulent information (Mishra et al., 2023).

A multi-pronged approach is necessary for the successful development of an effective false news detection system. Algorithms like Support Vector Machine (SVM), Nearest Neighbour, Random Forest, Naive Bayes, and Logistic Regression are employed in this approach. Applying these models to a variety of datasets gathered from various platforms is necessary to create a robust system that can generalize across various types of news articles.(Sharma et al., 2017).

The first step in this process is to gather datasets from different sources. Social networking networks and online news portals might offer datasets for this. Each dataset presents unique challenges and features which ensure a comprehensive evaluation of the models' applications in real-world scenarios.

Text preparation has a crucial role in enhancing the models' capacity to distinguish bogus news. To do this, in order to extract valuable information from the text input, operations like encoding, stemming, and stop word removal must be carried out. Moreover, the textual data can be represented in a way suitable for machine learning models by using techniques like word embeddings of Term Frequency-Inverse Document Frequency (Aphiwongsophon et al., 2018).

Decision trees can be trained on several subsets of the dataset using an advanced ensemble learning method called Random Forest. The collective wisdom of these trees reduces overfitting and strengthens the model's robustness, making it perfect for spotting misleading information. Naive Bayes is a probabilistic classifier that assumes feature independence. It is a valuable addition to the model collection since, despite its simplicity, it often achieves good results in text categorization tasks. Nearest Neighbour uses similarity measures to classify instances. By considering the proximity of data points in a feature space, it can identify trends in the dataset and identify fake news more effectively. A good linear model for tasks requiring binary classification is logistic regression. It can be used to simulate the chance that a news piece is fraudulent and produce results that are easy to grasp. The most effective hyperplane to divide classes is found using the complex classification method known as Support Vector Machine (SVM). Text classification makes good use of SVM's high-dimensional feature space handling capabilities.

These models are systematically applied in order to develop a flexible and precise false news detection system. This is due to the fact that they use preprocessed datasets from various sources. It is necessary to consistently assess these models and make refinements where necessary. This will help better the detecting of changing patters across various datasets. This in turn will assist in identifying the spread of false information.

Literature Review

The potential for fake news to upset the balance of authenticity in the news ecosystem is one of its noteworthy effects. On websites like Facebook, fake news occasionally surpassed real mainstream news during the 2016 U.S. elections. This false information flow casts doubt on how valid news sources are and erodes people's confidence in the information they come across. Intentionally influencing customers to embrace skewed or incorrect ideas is another purpose of fake news, which propagandists frequently employ to spread political messages or sway public opinion (Khan et al., 2017). An example of the purposeful use of fake news for political benefit is Russia's use of social bots and phoney accounts to propagate misleading information.

Moreover, the purpose of false news is to persuade viewers to embrace inaccurate or biased opinions. Most reporters who spread false information under the disguise of fake news are propagators attempting to sway public opinion. Some say that Russia uses social bots and fake accounts to spread false information, which is an example of fake news propagators.

Developing methods for automatically detecting false information is crucial to reducing the harmful impacts of fake news. The main place to accomplish this is on social media. Because it is convenient and inexpensive to manufacture, fake news is produced in large quantities, which makes automated identification essential. Machine learning and artificial intelligence (AI) have emerged as viable remedies for this issue (Villela et al., 2023).

Research evaluating several approaches to false news detection have evaluated the performance of each algorithm across several datasets. A couple of the challenges include the need for trustworthy models that perform well with various types of news and dataset bias. Moreover, the employment of advanced pre-trained language models such as BERT, ELECTRA, and ELMo has shown promise in improving detection accuracy (Shu et al., 2017). One way to detect how effective machine learning algorithms are can be done by thoroughly analyzing them. This covers both conventional (SVM, LR, Decision Tree, Naive Bayes, k-NN) and deep learning (CNN, LSTM, Bi-LSTM, C-LSTM, HAN, Conv-HAN) models.

As a global platform for real-time discussion, social media platforms have been instrumental in the dissemination of false information. Because users can search for tweets about particular subjects using the platform's streaming API, it is an invaluable tool for researchers examining the mechanics of fake news on social media platforms (Ahmed et al., 2020). The scientific community has realized that in order to categorize news as authentic or fraudulent, real-time data extraction from social media platforms and analyzing them later using machine learning algorithms are required.

Methodology

The detection of fake news in this project was done in three steps. Initially, the data was collected and cleaned, before being preprocessed in the next steps. Finally, the data was sent to be analyzed to produce the results which will be explained later.

Data Collection and Cleaning

The BuzzFeedNews and FakeNews datasets (https://github.com/BuzzFeedNews/2016-10-facebook-fact-check/tree/master/data) were the two separate datasets that were gathered and refined in the initial phase of the research.182 rows and 12 columns made up the original BuzzFeedNews dataset, which was taken from their GitHub repository. Since this dataset is smaller than most, preparation and cleaning were done to make sure it was suitable for further research. The primary objective during this phase was to distill relevant information for the study, leading to the final selection of five critical columns: title, text, source, contain_image, and news_type.

Similarly, the FakeNews dataset, sourced from Kaggle, (https://www.kaggle.com/mrisdal/fake-news) posed a more substantial challenge due to its larger scale, consisting of 12,999 rows and 20 columns. To streamline this dataset, the cleaning process involved standardizing column names, addressing missing or irrelevant entries, and eliminating unnecessary features. To make the dataset more manageable and conducive to analysis, a strategic decision was made to downsize it by randomly shuffling and selecting a

representative subset of 4,000 rows. This not only reduced computational complexity but also retained the diversity of the dataset.

In addition to this, both of these datasets did not have a contain_image column. This column was generated by comparing the value of the image column in regards to the contain_image column. This process was repeated twice to derive two contain image columns for both datasets, i.e. the BuzzFeed and the Kaggle datasets.

Another shortcoming of the Kaggle dataset was that it did not contain a news_type column in its dataset. This column was generated on the basis of spam scores. This was done by checking if the spam score is greater than or less than 0 after subtracting the actual value from 0.5. If greater the news was regarded as fake, else it was counted as real news.

spam_score = spam_score - 0.5

if spam_score > 0 Fake, else Real

Regarding fake news detection, the chosen columns in both datasets were important. The news articles' text was supplied by the columns labelled "title" and "text," and the news's original source was indicated by the column labelled "source." Images were noted in the 'contain_image' column, and the 'news_type' column stated if the news was classified as true or fraudulent. These columns functioned as the fundamental elements of the analysis that followed, guaranteeing that pertinent data was kept and irrelevant details were eliminated.

This phase's last step involved combining the adjusted FakeNews dataset and the cleaned BuzzFeedNews dataset to create a consolidated dataset with 4,182 rows. Combining data from many sources gave a larger set of analysis to form a perspective on the news articles. In the later phases of the project, the generated dataset including the five chosen columns was used as the foundation for additional analytics, text preparation, and machine learning model training.

Analytics and Text Preprocessing

Performing analytics on the combined dataset was a critical step that came after the datasets were joined. Exploratory data analysis (EDA), which includes a variety of methods to gain a deeper understanding of the dataset's properties, was probably a part of this analytics phase. Some of the examples of the data analysis include examining statistics, finding trends and feature distribution. The information from this phase would direct further decisions and preprocessing steps.

The purpose of the following text preprocessing steps was to improve the textual data's quality and relevance for machine learning analysis. Every stage of text preprocessing has a distinct function in converting unprocessed text into a format that may be used to train machine learning models.

- **Lowercasing:** Lowercasing is a common practice to ensure uniformity in text data. It converts all text to lowercase, preventing the model from treating words with different cases as distinct entities. This step reduces redundancy and improves consistency.

- **Removing Numbers:** The removal of numbers from the text is essential to eliminate numerical information that may not contribute to the classification of news types. It simplifies the text, focusing on linguistic patterns rather than numerical values.
- **Removing Punctuation:** Punctuation removal streamlines the text further by eliminating symbols that do not carry significant meaning in natural language processing. This step aids in reducing noise and simplifying the language representation.
- **Removing Special Characters:** The removal of special characters while retaining alphanumeric characters and spaces ensures that only relevant textual information is considered. This step aims to create a cleaner, more standardized representation of the text.
- **Removing English Stopwords:** Stopwords, common words like "and," "the," and "is," were removed to focus on more meaningful terms. Eliminating stopwords reduces noise in the data and emphasizes words with higher semantic value.
- **Stemming:** The application of stemming involves reducing words to their root forms, aiding in normalization and capturing core meanings. The Porter Stemmer from NLTK is a popular choice for this task.
- **Tokenization:** Tokenization is a process that breaks down the text into individual units, such as words or phrases. NLTK's Whitespace Tokenizer was used for this, creating a structured format that are effectively processed by machine learning models.

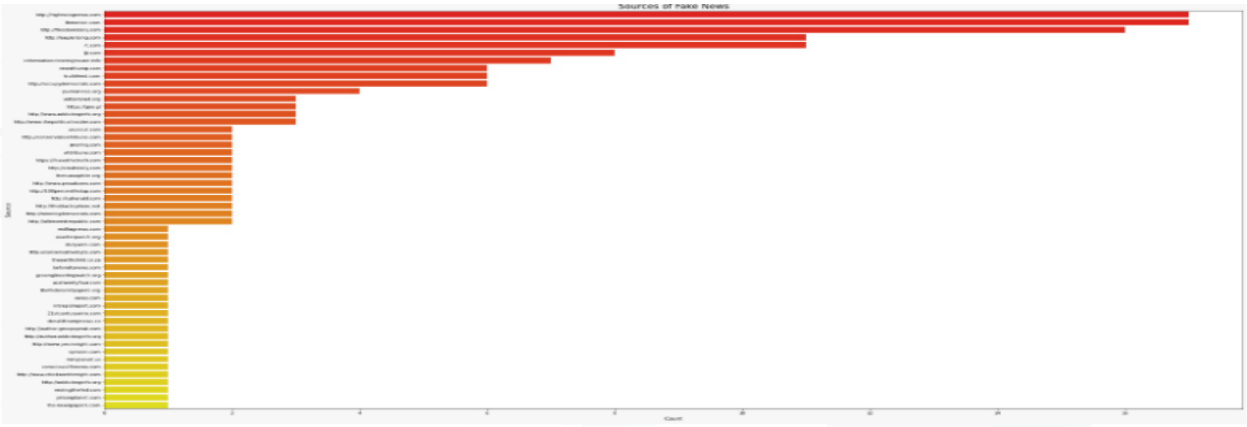


Figure 1 Sources of Fake News (i)

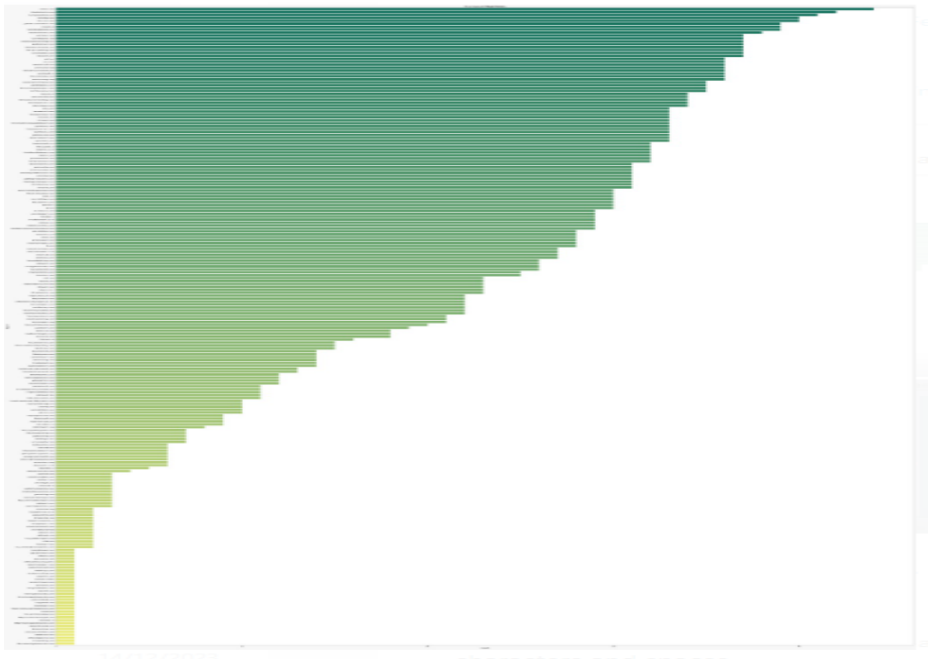


Figure 2 Sources of Fake News (ii)

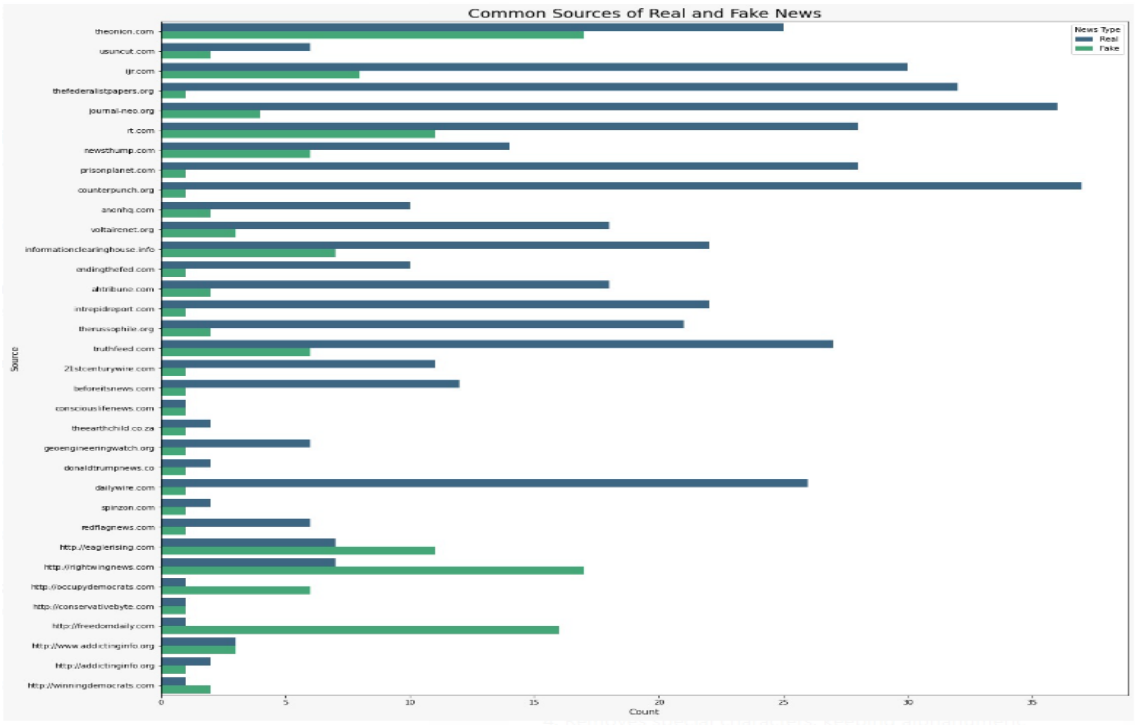


Figure 3 Sources of Fake News (iii)

The sources of the fake news as shown in Figures 1,2 and 3 were obtained from the datasets as provided by BuzzFeed and Kaggle. These sources were then analyzed to obtain the results as shown below.

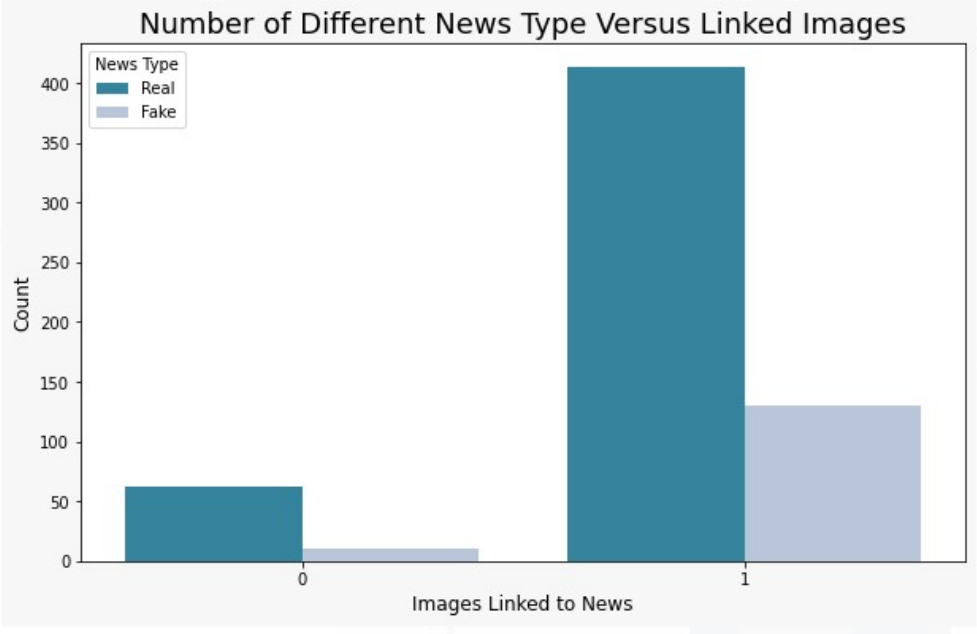


Figure 4 Number of Different News Type Versus Linked Images

Figure 4 shows how similar were the linked images to the news types, which can be an important demographic when detecting fake news.

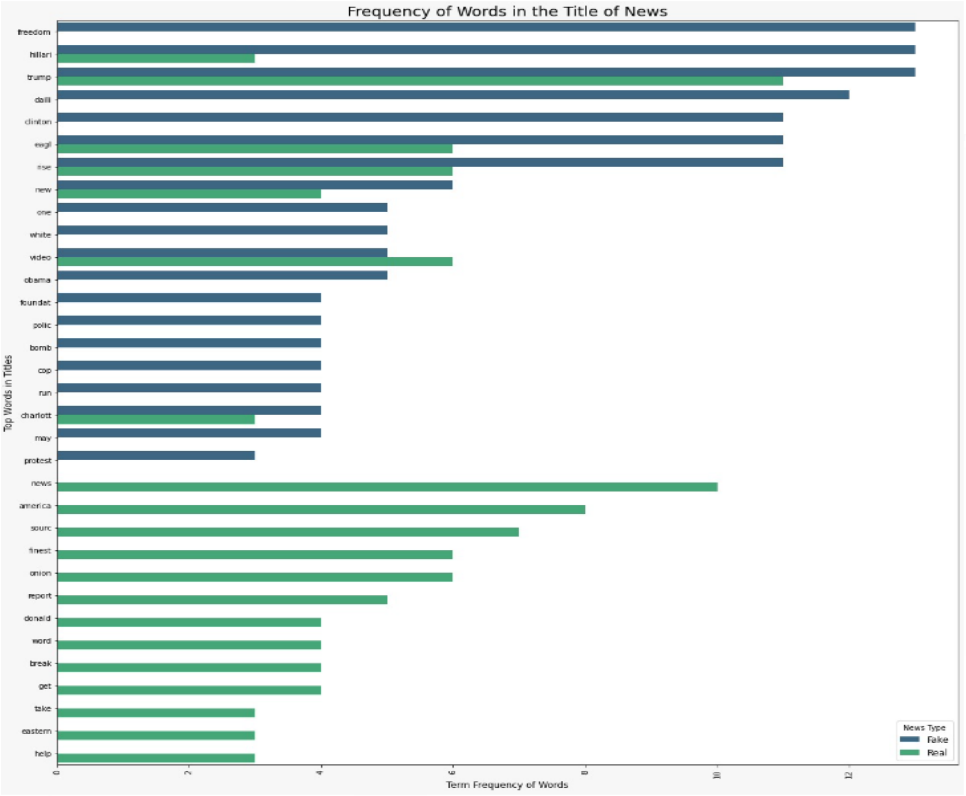


Figure 5 Word Frequency in News Text (i)

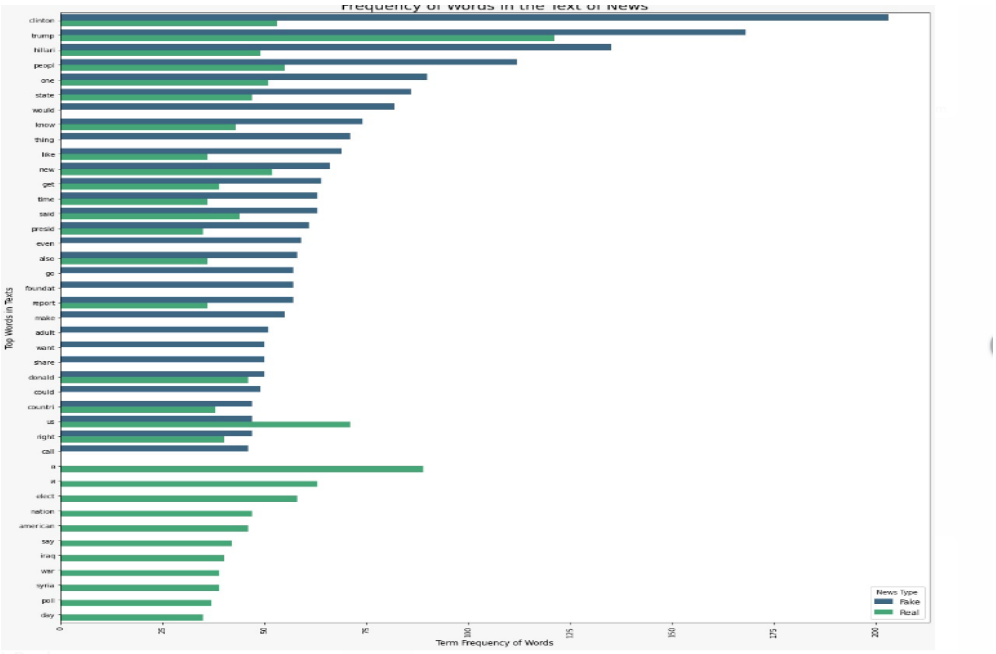


Figure 6 Word Frequency in News Text (ii)

Figure 5 and 6 show how frequently certain words appeared in News Titles, through which it became easier to differentiate between fake and real news.

These preprocessing steps collectively contribute to refining the textual data, ensuring that the machine learning models receive input that is standardized, relevant, and conducive to accurate classification. This meticulous preparation sets the stage for feature extraction and model training in subsequent phases of the analysis.

Features Extraction, Model Training and Results

The combined dataset was subjected to feature extraction, model training, and evaluation in the last stage of the investigation. The objective was to create a reliable system that could identify, using specific textual characteristics, between news that was authentic and that was not. The procedure was choosing the news_type column to be the target variable (Y) and the text column to be the feature variable (X).

Two feature extraction programmes, CountVectorizer and TfidfTransformer, were used to convert the textual data into a format appropriate for machine learning models.

1. **CountVectorizer:** This straightforward but effective tool creates a matrix of token counts from several text sources. A document is represented by each row in the matrix, and a unique token from the corpus is represented by each column. The values in the matrix show how frequently each token appears in the corresponding documents. By converting language into numerical characteristics, machine learning algorithms are able to handle and evaluate textual input more efficiently.

2. **TfidfTransformer:** Frequency-Inverse Document Frequency is a statistic that can be used to determine a word's significance inside a document in relation to a group of documents. TfidfTransformer is used to further enhance the feature representation after CountVectorizer. Tokens

are given weights according to how frequently they occur within a particular text and how rarely they occur within the corpus as a whole. This normalization highlights the value of unique terms in each document while lessening the influence of common ones.

After the extraction of features, the data was processed and prepared for model training. Although the precise algorithms were not stated, five supervised machine learning classification algorithms were used. These algorithms were used to identify trends in the text data's feature representations and forecast the kind of news—genuine or fraudulent.

The labelled dataset, in which each instance was connected to a particular news genre, was presented to the algorithms as part of the model training process. During this training phase, the algorithms discovered the patterns that were present in the text and how they related to the different kinds of news.

Lastly, performance indicators were used to assess the trained models, with accuracy being emphasised as the main metric. After that, a number of techniques, including Support Vector Machine, Random Forest, Naïve Bayes, Nearest Neighbour, and Logistic Regression, were used to assess the models. Next, the algorithms were ranked based on F-1 support, accuracy, precision, and recall. With an 81% success rate in properly detecting news genres, the Support Vector Machine (SVM) model proved to be the most accurate among the tested models. Although Logistic Regression was the least dependable approach, Naive Bayes also produced extremely accurate findings (Jain et al., 2019). This shows that SVM performed better than other algorithms in differentiating between bogus and real news in the provided dataset, based on the chosen textual features and the preprocessing processes. The accuracy statistic highlights the SVM model's dependability in this situation by displaying the percentage of successfully classified occurrences.

Future Research

When detecting and analyzing if news was fake or real, an assumption was made that all news over 0.5 spam score are real, whilst the others are fake. This assumption led to the majority of the samples being skewed towards the real samples. If some other factor, other than 0.5 would be taken, it would have led to completely different results. For example, using a higher factor would mean more fake results, while vice versa, a lower factor would mean more real results. To counter this some other metric besides spam score can also be used in order to derive results.

Another assumption made was that the independent feature (X) in this study was ‘text’, while the target variable (Y) was news_type. In the future, further research can be carried out on the same data by changing the independent feature to title or a combination of ‘text’ and ‘title’. This would yield different results, and the average of all the studies would mean a more accurate result.

Thirdly, the model used 4000 records for computation of the results. This had to be done due to the restriction of the computational power being used. For more accurate results, a stronger computational

algorithm can be used to compute even larger models (Yuan et al., 2023). Another compromise made within the data set was that all of the models were created using SciKit Learn. The disadvantage of using this library means that the data set was not computed in GPU, due to hardware restrictions. A stronger library that generates models that use GPU's (PyTorch or TensorFlow) can be used in order to broaden the scope of the study and to obtain more realistic results.

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